

Student Name _____

Class _____

2017

Higher School Certificate

Term 4 T1

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 90 minutes
- NESA approved calculators may be used
- Write using black pen
- Marks may not be awarded for careless or badly arranged work
- An HSC Reference Sheet is provided
- All necessary working should be shown

Write your student number at the top of every page

Total marks – 60

This examination consists of 6 questions worth 10 marks each.

The answers to all questions are to be returned in separate bundles clearly labelled Question 1, Question 2, etc.

This paper MUST NOT be removed from the examination room.

Question 1 (10 Marks)

(a) Find the coefficient of the x^4 term in the expansion of $\sum_{k=0}^6 \binom{6}{k} (x^2)^{6-k} (3)^k$ 2

(b) Evaluate x and y if $(\sqrt{3} - 1)^5 = x + y\sqrt{3}$ 2

(c) Use the substitution $u = t^2 + 2$ to find $\int t^3 \sqrt{t^2 + 2} dt$ 3

(d) If $\alpha = \cos^{-1} x$ and $\beta = \cos^{-1} 3y$ show that $\cos(\cos^{-1} x - \cos^{-1} 3y) = 3xy + \sqrt{(1-x^2)(1-9y^2)}$ 3

Question 2 (10 Marks) Start a new sheet of paper

(a) Find the constant term in the expansion $(5 + x)^4 \left(1 - \frac{2}{x}\right)^3$ 2

(b) Differentiate $x^2 \cos^{-1}(3x)$ 2

(c) Find the exact value of $\sin^{-1}(\sin \pi)$ 1

(d) Find the gradient to the curve $y = \tan^{-1}(\sin x)$ at $x = \pi$ 2

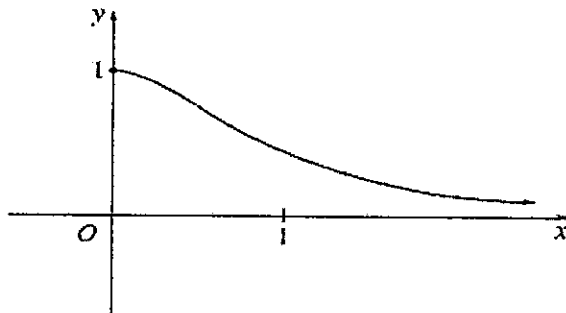
(e) Find the general solution of the equation $2 \cos 3x \sin 4x + 2 \cos 3x - \sin 4x - 1 = 0$ 3

Question 3 (10 Marks) Start a new sheet of paper

- (a) The coefficients of x and x^2 in the expansion of $(1 + ax)^n$ are -20 and 180 respectively.
Find the values of a and n 3
- (b) Find the exact value of $\cos(2 \cos^{-1} x + \sin^{-1} x)$ when $x = \frac{1}{5}$ 3
- (c) (i) Find the domain and range of the function $f(x) = 2 \cos^{-1}(1 - x)$ 2
- (ii) Sketch the graph of the function $f(x) = 2 \cos^{-1}(1 - x)$
Label clearly the coordinates of the end points. 2

Question 4 (10 Marks) Start a new sheet of paper

- (a) This is the graph of $y=f(x)$



- Copy the graph onto your answer and draw the graph of $y = f^{-1}(x)$ on the same set of axes 2
- (b) What is the inverse function of $y = \frac{1}{4-x}$? 1
- (c) If $f(x) = 2 \cos^{-1}\left(\frac{x}{\sqrt{2}}\right) - \sin^{-1}(1 - x^2)$
- (i) Find $f'(x)$ 2
- (ii) Hence or otherwise, show that $2 \cos^{-1}\left(\frac{x}{\sqrt{2}}\right) - \sin^{-1}(1 - x^2) = \frac{\pi}{2}$ 2
- (d) Water is pumped out of a 10 000 litre storage container through a valve such that the water is flowing out of the storage container at the rate of $2.4t$ litres per second ($t \geq 0$).
- (i) Show that if the tank was initially full, then the volume of water in the tank is given by $V = 10\,000 - 1.2t^2$. 2
- (ii) How long before the tank is only 30% full? Answer to the nearest second. 1

Question 5 (10 Marks) Start a new sheet of paper

- (a) Find the exact area bounded by the curve $y = 3 \sin^{-1}(2x)$, the x -axis and the line $x = \frac{1}{2}$ 4
- (b) (i) Use the substitution $y = \sqrt{x}$ to find $\int \frac{dx}{\sqrt{x(1-x)}}$ 2
- (ii) Use the substitution $z = x - \frac{1}{2}$ to find another expression for $\int \frac{dx}{\sqrt{x(1-x)}}$ 2
- (iii) Use the results of parts (i) and (ii) to express $\sin^{-1}(2x - 1)$ in terms of $\sin^{-1} \sqrt{x}$ for $0 < x < 1$ 2

Question 6 (10 Marks) Start a new sheet of paper

- (a) Let $(3 + 2x)^{20} = \sum_{r=0}^{20} a_r x^r$.
- i) Write an expression for a_r 1
- ii) Show that $\frac{a_{r+1}}{a_r} = \frac{40-2r}{3r+3}$ 2
- iii) Hence find the greatest coefficient in the expansion of $(3 + 2x)^{20}$ 2
- (b) Find the exact value of $\int_1^2 \frac{4}{\sqrt{4-x^2}} dx$ 2
- (c) Find $\frac{d}{dx} (\tan^{-1} \frac{x}{3})^2$ and hence find the exact value $\int_0^{\sqrt{3}} \frac{\tan^{-1}(\frac{x}{3})}{9+x^2} dx$ 3

END OF EXAMINATION

2017 Ext 1 Y. 11/12 Yrly Q1

$$a) x^{12-2k} = x^4$$

$$\therefore k=4$$

$$\text{coeff of } x^4 \text{ is } \binom{6}{4} 3^4 = \underline{1215}$$

must have 1215
to get 2nd mark

$$b) (\sqrt{3}-1)^5 = (\sqrt{3})^5 - 5(\sqrt{3})^4 + 10(\sqrt{3})^3 - 10(\sqrt{3})^2 + 5\sqrt{3} - 1$$

$$= 9\sqrt{3} - 45 + 30\sqrt{3} - 30 + 5\sqrt{3} - 1$$

$$= 44\sqrt{3} - 76$$

$$\therefore x = -76 \quad y = 44$$

must have
1 -5 10 70 5 -1
to get 1st mark

$$c) \frac{1}{2} \int t^2 \sqrt{t^2+2} (2t dt) \quad u = t^2+2$$

$$du = 2t dt$$

$$= \frac{1}{2} \int (u-2) \sqrt{u} du$$

$$= \frac{1}{2} \int u^{3/2} - 2u^{1/2} du$$

$$= \frac{1}{5} u^{5/2} - \frac{2}{3} u^{3/2} + C$$

$$= \frac{1}{5} (t^2+2)^{5/2} - \frac{2}{3} (t^2+2)^{3/2} + C$$

$$d) \text{ let } \alpha = \cos^{-1} x, \quad \beta = \cos^{-1} 3y$$

$$\therefore \cos \alpha = x \quad \cos \beta = 3y$$

$$\sin^2 \alpha + \cos^2 \alpha = 1, \quad \sin^2 \beta + \cos^2 \beta = 1$$

$$\sin^2 \alpha = 1 - \cos^2 \alpha \quad \sin^2 \beta = 1 - \cos^2 \beta$$

$$\sin \alpha \quad 0 \leq \alpha, \beta \leq \pi$$

$$\therefore \sin \alpha = \sqrt{1-x^2}, \quad \sin \beta = \sqrt{1-9y^2}$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$= 3xy + \sqrt{1-x^2} \sqrt{1-9y^2}$$

$$\cos(\cos^{-1} x - \cos^{-1} 3y) = 3xy + \sqrt{(1-x^2)(1-9y^2)}$$

Mathematics Ex+1 Question 2.

Suggested Solutions

Marks

Marker's Comments

$$\begin{aligned} \text{a) General Term for } (5+x)^4 \text{ is } {}^4C_r 5^{4-r} x^r \\ \text{General Term for } (1-\frac{2}{x})^3 \text{ is } {}^3C_k (1)^{3-k} (-2x^{-1})^k \\ = {}^3C_k (-2)^k (x^{-k}) \end{aligned}$$

①

$$\therefore \text{General Term for product is } {}^4C_r 5^{4-r} x^r {}^3C_k (-2x)^{-k}$$

$$\text{For constant: } x^r \cdot x^{-k} = 1 \\ r=k$$

$$\therefore \text{When } r=k=0: {}^4C_0 5^4 \times {}^3C_0 = 625$$

$$r=k=1, {}^4C_1 5^3 \times {}^3C_1 (-2) = -3000$$

$$r=k=2, {}^4C_2 5^2 \times {}^3C_2 (-2)^2 = 1800$$

$$r=k=3, {}^4C_3 5 \times {}^3C_3 (-2)^3 = -160$$

$$\therefore \text{Coefficient is } 625 - 3000 + 1800 - 160 \\ = -735$$

①

or

Suggested Solutions	Marks	Marker's Comments
$a) (5+x)^4 \left(1 - \frac{2}{x}\right)^3$ $= \left(5^4 + {}^4C_1(5)^3x + {}^4C_25^2x^2 + {}^4C_35x^3 + x^4\right) \times$ $\left(1 - {}^3C_1\left(\frac{-2}{x}\right) + {}^3C_2\left(\frac{2}{x}\right)^2 - \left(\frac{2}{x}\right)^3\right)$ $\therefore$ constant term is : $5^4 + {}^4C_1(5)^3 {}^3C_1(-2) + {}^4C_25^2 {}^3C_2(-2)^2 + {}^4C_35 \times (-2)^3$ $= -735$		①
$b) \frac{d}{dx} [x^2 \cos^{-1}(3x)]$ $= x^2 \frac{-1}{\sqrt{1-9x^2}} \times 3 + \boxed{2x \cos^{-1}(3x)}$ $= 2x \cos^{-1}(3x) - \boxed{\frac{3x^2}{\sqrt{1-9x^2}}}$		① ①
$c) \sin^{-1}(\sin \pi)$ $= \sin^{-1}(0)$ $= 0$		①
$d) \frac{dy}{dx} = \frac{1}{1+\sin^2 x} \times \cos x$ $= \frac{\cos x}{1+\sin^2 x}$ When $x = \pi$, $\frac{dy}{dx} = \frac{-1}{1+0} = -1$	<u>Note:</u> $1 + \sin^2 x \neq \cos^2 x$	① ①

Mathematics Ext1 Question 2.

Suggested Solutions	Marks	Marker's Comments
e) $2\cos 3x \sin 4x + 2\cos 3x - \sin 4x - 1 = 0$ $2\cos 3x (\sin 4x + 1) - (\sin 4x + 1) = 0$ $(2\cos 3x - 1)(\sin 4x + 1) = 0$ $\therefore 2\cos 3x - 1 = 0 \quad \text{or} \quad \sin 4x + 1 = 0$ <u>1st</u>: $\cos 3x = \frac{1}{2} \quad (n \in \mathbb{Z})$ $3x = 2n\pi \pm \frac{\pi}{3}$ $x = \frac{2n\pi}{3} \pm \frac{\pi}{9}$ <u>2nd</u> $\sin 4x = -1$ $4x = n\pi + (-1)^n \left(-\frac{\pi}{2}\right) \quad (n \in \mathbb{Z})$ $x = \frac{n\pi}{4} + (-1)^n \left(-\frac{\pi}{8}\right)$ <u>1st Mark</u> — Factorise <u>2nd Mark</u> — 1st correct general solution <u>3rd Mark</u> — 2nd correct general solution and defining $n \in \mathbb{Z}$.		other general solutions $\rightarrow \frac{n\pi}{4} + (-1)^n \frac{3\pi}{8},$ $\frac{n\pi}{2} + \left(-\frac{\pi}{8}\right)$ $\frac{n\pi}{2} + \left(\frac{3\pi}{8}\right)$

MATHEMATICS Extension 1 : Question 3

Suggested Solutions

Marks

Marker's Comments

$$a) (1+ax)^n = 1 + {}^n C_1 ax + {}^n C_2 a^2 x^2 + \dots$$

$$\text{Coefficient of } x: na = -20$$

$$\text{Coefficient of } x^2: {}^n C_2 a^2 = 180$$

$$na = -20$$

$$\frac{n! a^2}{2!(n-2)!} = 180$$

$$n(n-1)a^2 = 360$$

$$a^2 n^2 - a^2 n = 360$$

$$(-20)^2 - a(-20) = 360$$

$$400 + 20a = 360$$

$$20a = -40$$

$$a = -2$$

$$-2n = -20$$

$$n = 10$$

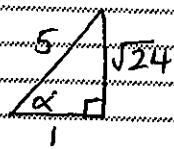
$$b) \cos(2\cos^{-1}x + \sin^{-1}x), x = 1/5$$

Method 1:

$$\text{Let } \alpha = \cos^{-1} 1/5$$

$$\cos \alpha = 1/5$$

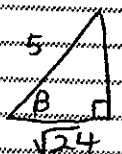
$$\therefore 0 < \alpha < \pi/2$$



$$\text{Let } \beta = \sin^{-1} 1/5$$

$$\sin \beta = 1/5$$

$$\therefore 0 < \beta < \pi/2$$



$$\therefore \sin \alpha = \frac{2\sqrt{6}}{5}$$

$$\cos \beta = \frac{2\sqrt{6}}{5}$$

$$\cos(2\alpha + \beta) = \cos 2\alpha \cos \beta - \sin 2\alpha \sin \beta$$

$$= (\cos^2 \alpha - \sin^2 \alpha) \cos \beta$$

$$= \left(\frac{1}{25} - \frac{24}{25}\right) \frac{2\sqrt{6}}{5} - 2 \times \frac{2\sqrt{6}}{5} \times \frac{1}{5} \times \frac{1}{5}$$

$$= \frac{-23}{25} \times \frac{2\sqrt{6}}{5} - \frac{4\sqrt{6}}{125}$$

$$= \frac{-50\sqrt{6}}{125} = \frac{-2\sqrt{6}}{5}$$

coefficients with ${}^n C_2$ in factorial notation

linear equation in a or n

a and n

Note:

$$\begin{aligned} \cos 2\alpha &= 2\cos^2 \alpha - 1 \\ &= 1 - 2\sin^2 \alpha \\ &= \cos^2 \alpha - \sin^2 \alpha \end{aligned}$$

$\sin \alpha$ and $\cos \beta$

expansion with double angles

answer

MATHEMATICS Extension 1 : Question 3

Suggested Solutions

Marks

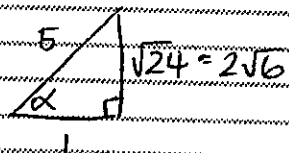
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Method 2:

$$\begin{aligned} & \cos(2\cos^{-1}x + \sin^{-1}x) \\ &= \cos(\cos^{-1}x + \cos^{-1}x + \sin^{-1}x) \\ &= \cos(\cos^{-1}x + \pi/2) \quad \text{since } \cos^{-1}x + \sin^{-1}x = \pi/2 \end{aligned}$$

$$\begin{aligned} \text{Let } \alpha &= \cos^{-1}x \\ \cos \alpha &= x \\ &= 1/5 \end{aligned}$$

$$\therefore 0 < \alpha < \pi/2$$



$$\therefore \sin \alpha = 2\sqrt{6}/5$$

$$\begin{aligned} & \cos(\cos^{-1}x + \pi/2) \\ &= \cos(\alpha + \pi/2) \end{aligned}$$

$$= \cos \alpha \cos \pi/2 - \sin \alpha \sin \pi/2$$

$$= \cos \alpha \times 0 - \sin \alpha \times 1$$

$$= -2\sqrt{6}/5$$

Note : marks were not deducted for omission of domain of α and β , however the best answers showed why α and β are acute.

Students should consider what their answers would have been had x been negative.

1

$\cos^{-1}x + \sin^{-1}x = \pi/2$
and
 $\cos(\cos^{-1}x + \pi/2)$

1

$\sin \alpha$

1

answer

MATHEMATICS Extension 1 : Question 3

Suggested Solutions

Marks

Marker's Comments

c) $f(x) = 2\cos^{-1}(1-x)$

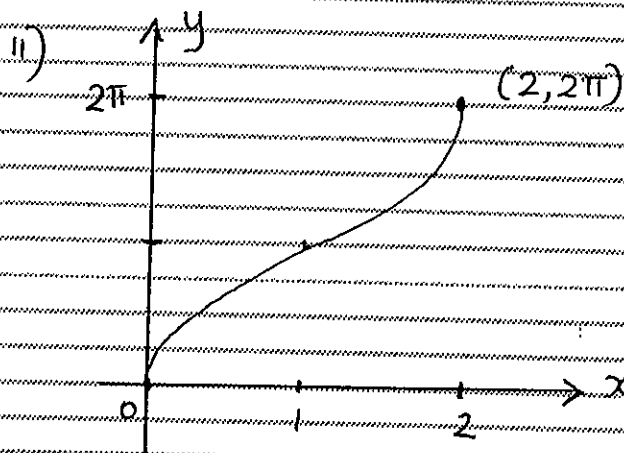
i) Domain: $-1 \leq 1-x \leq 1$
 $1 \geq x-1 \geq -1$
 $2 \geq x \geq 0$

Range: $0 \leq y \leq \pi$
 $\frac{\pi}{2}$

$0 \leq y \leq 2\pi$

Domain: $\{x: 0 \leq x \leq 2\}$

Range: $\{y: 0 \leq y \leq 2\pi\}$



1 domain

1 range

1 correct shape

1 endpoints

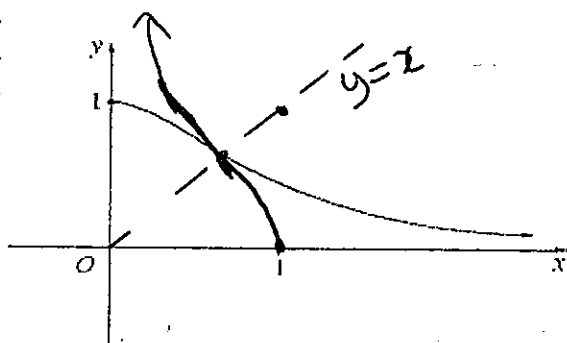
MATHEMATICS Extension 1 : Question 4...

Suggested Solutions

Marks

Marker's Comments

a)



2

1 for evidence of reflection

in $y = x$

1 for correct graph

$$b) \quad y = \frac{1}{4-x}$$

$$x = \frac{1}{4-y}$$

$$x(4-y) = 1$$

$$xy = 4x - 1$$

$$y = \frac{4x-1}{x}$$

$$c) \quad f(x) = 2 \cos^{-1} \frac{x}{\sqrt{2}} - \sin^{-1}(1-x^2)$$

$$f'(x) = \frac{-2}{\sqrt{2-x^2}} - (-2x) \frac{1}{\sqrt{1-(1-x^2)^2}}$$

$$= \frac{-2}{\sqrt{2-x^2}} + \frac{2x}{\sqrt{2x^2-x^4}}$$

$$= \frac{-2}{\sqrt{2-x^2}} + \frac{2x}{x\sqrt{2-x^2}} \quad \text{if } x > 0$$

$$= 0$$

1

well done

had to be

written in

form $y =$ to gain mark

1 mark for

correctly differentiated

MATHEMATICS Extension 1 : Question 4...

Suggested Solutions

Marks

Marker's Comments

(ii) Since $f'(x) = 0 \quad x > 0$

$f(x) = c$ or $f(x)$ is a horizontal line
so testing a point

$$f(1) = 2\cos^{-1}\frac{1}{\sqrt{2}} - \sin^{-1}1$$

$$= 2 \times \frac{\pi}{2} - \frac{\pi}{2}$$

$$= \frac{\pi}{2}$$

$$\therefore f(x) = 2\cos^{-1}\left(\frac{x}{\sqrt{2}}\right) - \sin^{-1}(1-x)$$

$$= \frac{\pi}{2} \quad 0 < x < \sqrt{2}$$

(d) (i) $\frac{dv}{dt} = -2.4t$

$$v = -1.2t^2 + c$$

when $t=0 \quad v=10,000$

$$\therefore c = 10,000$$

$$v = -1.2t^2 + 10,000$$

(ii) $30\% \times 10,000 = 3000$

$$1.2t^2 = 7000$$

$$t = \sqrt{\frac{7000}{1.2}}$$

$$= 76 \text{ seconds}$$

accepted (77s)

NB: A straight line is not a horizontal

students lost the negative

$$-30\% \neq \frac{1}{3}$$

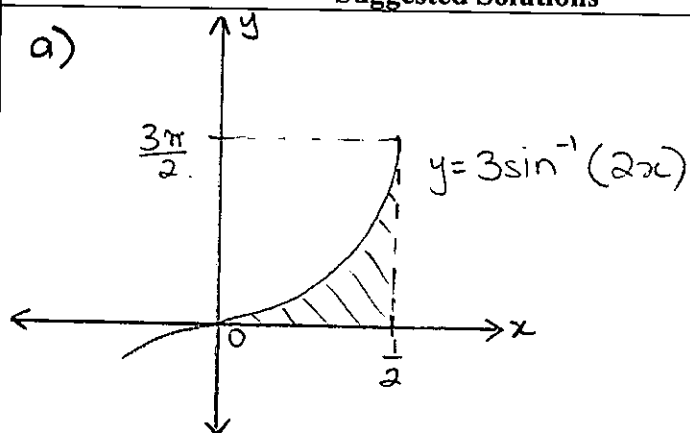
MATHEMATICS Extension 1: Question 5

Suggested Solutions

Marks

Marker's Comments

a)



$$y = 3 \sin^{-1}(2x)$$

$$\frac{y}{3} = \sin^{-1}(2x)$$

$$2x = \sin \frac{y}{3}$$

$$x = \frac{1}{2} \sin \frac{y}{3}$$

$$\begin{aligned} \text{Area} &= \frac{3\pi}{2} \times \frac{1}{2} - \int_0^{\frac{3\pi}{2}} \frac{1}{2} \sin \frac{y}{3} dy \\ &= \frac{3\pi}{4} - \frac{1}{2} \left[-3 \cos \frac{y}{3} \right]_0^{\frac{3\pi}{2}} \\ &= \frac{3\pi}{4} - \frac{1}{2} \left(-3 \cos \frac{3\pi}{2} + 3 \cos 0 \right) \\ &= \frac{3\pi}{4} - \frac{1}{2} (0 + 3) \\ &= \frac{3\pi}{4} - \frac{3}{2} \end{aligned}$$

b) i) $y = \sqrt{x} \Rightarrow x = y^2$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x}}$$

$$\therefore 2dy = \frac{dx}{\sqrt{x}}$$

$$\begin{aligned} \int \frac{dx}{\sqrt{x(1-x)}} &= \int \frac{2dy}{\sqrt{1-y^2}} \\ &= 2 \sin^{-1} y + c \\ &= 2 \sin^{-1} \sqrt{x} + c \end{aligned}$$

1 $\Rightarrow$ rectangle

1 $\Rightarrow$ correct upper and lower bounds and correct integral

1

1

1

need to substitute $\sqrt{x}$ back.

MATHEMATICS Extension 1: Question 5

Suggested Solutions

Marks

Marker's Comments

$$\text{ii) } z = x - \frac{1}{2} \Rightarrow x = z + \frac{1}{2}$$

$$\frac{dz}{dx} = 1$$

$$\therefore dz = dx$$

$$\int \frac{dx}{\sqrt{x(1-x)}} = \int \frac{dz}{\sqrt{(z+\frac{1}{2})(\frac{1}{2}-z)}}$$

$$= \int \frac{dz}{\sqrt{\frac{1}{4} - z^2}}$$

$$= \sin^{-1} 2z + c$$

$$= \sin^{-1} (2x-1) + c$$

1

1

iii) from (i) & (ii)

$$\sin^{-1} (2x-1) = 2 \sin^{-1} \sqrt{x} + c$$

when $x = \frac{1}{2}$, $0 < x < 1$

$$\sin^{-1} (0) = 2 \sin^{-1} \frac{1}{\sqrt{2}} + c$$

$$\therefore c = -\frac{\pi}{2}$$

$$\therefore \sin^{-1} (2x-1) = 2 \sin^{-1} \sqrt{x} - \frac{\pi}{2}$$

1

1

must have +c

use correct x
to find

$$c = -\frac{\pi}{2}.$$

$$6ai) \sum_{r=0}^{20} a_r x^r$$

$$(3+2x)^{20} = \sum_{r=0}^{20} {}^{20}C_r 3^{20-r} (2x)^r$$

$${}^{20}C_r 3^{20-r} (2)^r (x)^r$$

$$\therefore a_r = {}^{20}C_r 3^{20-r} 2^r \quad \text{--- (1)}$$

$$a_{r+1} = {}^{20}C_{r+1} 3^{20-(r+1)} (r+1)$$

$$= \frac{20!}{(r+1)!(20-(r+1))!} \times 3^{19-r} \times 2^{r+1}$$

$$= \frac{20!}{(r+1)!(19-r)!} \times 3^{19-r} \times 2^{r+1} \quad \text{--- (1)}$$

$$a_r = {}^{20}C_r 3^{20-r} 2^r$$

$$= \frac{20!}{r!(20-r)!} \times 3^{20-r} \times 2^r \quad \text{--- (1)}$$

$$\therefore \frac{a_{r+1}}{a_r} = \frac{\cancel{20!}}{(19-r)!(r+1)!} \times \frac{(20-r)! \cancel{(r)!}}{\cancel{20!}} \times \frac{3^{\cancel{19-r}}}{3^{20-r}} \times \frac{2^{r+1}}{\cancel{2^r}}$$

$$= \frac{20-r}{r+1} \times \frac{2}{3} = \frac{40-2r}{3r+3}$$

common error
not putting (r+1)
in brackets.
so students got
(21-r) which
is incorrect.

must be in
expanded form.

*. match up
so its easier
to cancel off

shown.

Suggested Solutions

Marks Awarded

Marker's Comments

iii) for greatest coefficient (G.C.)

$$\frac{a_{r+1}}{a_r} > 1$$

$$\therefore \text{then } C_{r+1} > C_r$$

$$40 - 2r > 3r + 3$$

$$37 > 5r$$

$$r < \frac{37}{5} \text{ or } 7\frac{2}{5} \text{ or } 7.4 \text{ --- } \textcircled{1}$$

check.

$$\text{when } r=7 \quad a_8 > a_7$$

$$r=6 \quad a_7 > a_6$$

$$r=0 \quad a_1 > a_0$$

$$\therefore a_8 > a_7 > a_6 > \dots > a_0.$$

$$\frac{a_{r+1}}{a_r} < 1 \text{ then } a_8 > a_7 > \dots > a_0$$

$$\frac{a_{r+1}}{a_r} > 1 \text{ holds true}$$

$$T_0 < T_1 < T_2 \dots < T_6 < T_7 < T_8 > T_9$$

$$\therefore \text{G.C.} = a_8$$

$$= {}^{20}C_8 \cdot 3^{12} \cdot 2^8$$

$$\text{or } 1.71 \times 10^{13}$$

} --- $\textcircled{1}$

Suggested Solutions

Marks Awarded

Marker's Comments

$$b) \int_1^2 \frac{4}{\sqrt{4-x^2}} dx$$

$$= 4 \int_1^2 \frac{1}{\sqrt{2^2-x^2}} dx$$

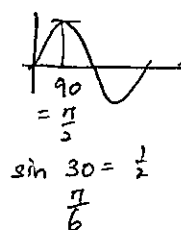
$$= 4 \left[\sin^{-1}\left(\frac{x}{2}\right) \right]_1^2$$

$$= 4 \left[\sin^{-1}(1) - \sin^{-1}\left(\frac{1}{2}\right) \right]$$

$$= 4 \left(\frac{\pi}{2} - \frac{\pi}{6} \right)$$

$$= \frac{4\pi}{3}$$

(1)



(1)

$$c) \frac{d}{dx} \left[\tan^{-1}\left(\frac{x}{3}\right) \right]^2$$

$$\text{Let } y = \left(\tan^{-1}\left(\frac{x}{3}\right) \right)^2$$

$$\frac{dy}{dx} = 2 \left(\tan^{-1}\left(\frac{x}{3}\right) \right) \times \frac{3}{3^2+x^2}$$

$$= \frac{6 \tan^{-1} \frac{x}{3}}{9+x^2}$$

Hence,

$$\int_0^{\sqrt{3}} \frac{\tan^{-1}\left(\frac{x}{3}\right)}{9+x^2} dx \quad \left. \begin{array}{l} \text{compare the} \\ \text{two.} \end{array} \right\}$$

$$\therefore = \frac{1}{6} \left[\tan^{-1}\left(\frac{x}{3}\right)^2 \right]_0^{\sqrt{3}}$$

(1)

(1)

* Common error
 $\rightarrow$ not multiplying 2 by 3
 $\rightarrow$ not differentiating the inside.

MATHEMATICS Extension 1 : Question.....

4/4

Suggested Solutions

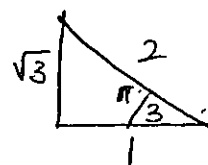
Marks Awarded

Marker's Comments

$$= \frac{1}{6} \left(\left(\tan^{-1} \frac{\sqrt{3}}{3} \right)^2 - \left(\tan^{-1} \frac{0}{3} \right)^2 \right)$$

$$= \frac{1}{6} \left(\frac{\pi}{6} \right)^2 - 0$$

$$= \frac{\pi^2}{216}$$



*common error

- not identifying the angle correctly
- not multiplying by $\frac{1}{6}$
- not squaring

1